

Verified bounds for positive cubic relaxation gaps

Abstract

Luedtke, Namazifar, and Linderoth proved that for bilinear functions with positive coefficients on the unit cube, the gap of the termwise relaxation, which uses the exact envelopes of the individual terms, is less than twice the gap of the convex hull of the graph. We study degree three. Let $R(3)$ be the supremum of the ratio of termwise gap to hull gap over multilinear polynomials of degree at most three with nonnegative coefficients, all finite nonnegative boxes, and all points with positive hull gap. We prove

$$2.1668 \approx \frac{1610000}{743033} \leq R(3) \leq \frac{31}{12} \approx 2.5833.$$

The upper bound uses one fixed mixture of three probability laws that preserves every variable mean and bounds every term at once. Its weights are optimal among mixtures of these three laws for a uniform termwise guarantee. The lower bound comes from an explicit integer-coefficient family and a rational Bernstein certificate. Its members lie on unit cubes and already for $m = 36$ have ratio above two, so some positive cubic polynomials on the unit cube have a larger ratio than every positive bilinear function there. The exact value of $R(3)$ remains open. An accompanying Lean project verifies both bounds, the optimality of the weights, and seven exact finite examples, all stated for the actual graph hulls.

1 Introduction

Global solvers for nonconvex problems usually relax each nonlinear term separately. For a multilinear function, the *termwise* relaxation replaces each monomial by its own convex and concave envelopes. The convex hull of the whole graph is stronger, but its standard description uses all vertices of the box [1]. We measure the loss by the ratio of the two vertical widths at a point.

Setting. Let $B = \prod_{i=1}^n [\ell_i, u_i]$ with $0 \leq \ell_i \leq u_i < \infty$, and let

$$f(x) = \sum_{e \in E} a_e m_e(x), \quad m_e(x) = \prod_{i \in e} x_i, \quad a_e \geq 0, \quad |e| \leq 3,$$

where E consists of distinct supports. Constant and affine terms are allowed. Write $\text{vex}_B g$ for the greatest convex underestimator of g on B , and $\text{cav}_B g$ for its least concave overestimator. The *hull gap* and the *termwise gap* are

$$\begin{aligned} H_B(f; x) &= \text{cav}_B f(x) - \text{vex}_B f(x), \\ T_B(f; x) &= \sum_{e \in E} [\text{cav}_B(a_e m_e)(x) - \text{vex}_B(a_e m_e)(x)]. \end{aligned}$$

Thus T_B uses the exact hull of each *original* term, and always $0 \leq H_B \leq T_B$. Define $R(3)$ as the supremum of T_B/H_B over this class, arbitrary finite dimension, and points with $H_B > 0$. Fixed coordinates, boundary points, and zero coefficients are allowed.

Prior work. Luedtke et al. [1] studied this comparison. For positive coefficients on a nonnegative box, the concave envelope of f is the sum of the concave envelopes of its terms, so all of the loss is on the convex side. For positive bilinear functions on the unit cube, where the termwise and McCormick relaxations coincide, they proved that T_B/H_B is less than two, with a factor depending on the chromatic number of the graph of the terms. They conjectured that for positive multilinear functions on nonnegative boxes the ratio is bounded by a constant independent of degree and dimension. Results for one fixed degree, such as ours, neither confirm nor refute this conjecture, which concerns all degrees at once. Degree three is the first case beyond their bilinear theorem. Our definitions use exact individual envelopes on each box; they are not identified with every recursive McCormick formulation, which can be weaker on a general positive box. We do not claim an exhaustive literature search for priority.

Results. Theorem 1 gives the two-sided bound on $R(3)$.

- *Upper bound* (Section 3). A single mixture (1) of three mean-preserving laws captures at least 12/31 of every bilinear and cubic term gap at once. This gives $T_B \leq (31/12)H_B$ on every finite nonnegative box.
- *Optimality of the weights* (Section 4). No fixed mixture of the same three laws guarantees more than 12/31 of every term gap.
- *Lower bound* (Section 5). The family (3) with positive integer coefficients, together with the scalar certificate of Lemma 2, gives ratios whose certified lower bounds converge to 1610000/743033. Already at $m = 36$ the certified bound without the slack δ exceeds two, which bounds every positive bilinear ratio on the unit cube [1].
- *Exact finite examples* (Section 6) and *formal verification* (Section 7).

The key tool is the probabilistic description of graph hulls in Section 2: on the unit cube, the convex envelope at x is the minimum of $\mathbb{E}f(X)$ over laws on binary vertices with mean x . The termwise gap lets each term choose its own law, while the hull gap requires one law for all terms.

Theorem 1. *For every polynomial and point above,*

$$T_B(f; x) \leq \frac{31}{12} H_B(f; x).$$

Moreover,

$$\frac{1610000}{743033} = \frac{4830000}{2229099} \leq R(3) \leq \frac{31}{12}.$$

2 Graph hulls and common laws

On the cube, separate affinity gives

$$\text{conv}\{(x, f(x)) : x \in [0, 1]^n\} = \text{conv}\{(v, f(v)) : v \in \{0, 1\}^n\}.$$

Consequently the lower and upper envelopes at x are the minimum and maximum of $\mathbb{E}f(X)$ over binary laws with $\mathbb{E}X_i = x_i$. These extrema are attained. For a nonempty support e , its individual endpoints are

$$\text{cav } m_e(x) = \min_{i \in e} x_i, \quad \text{vex } m_e(x) = \max \left(0, \sum_{i \in e} x_i - |e| + 1 \right).$$

The empty support is constant one. The common-threshold law $X_i = \mathbf{1}_{\{U \leq x_i\}}$, with U uniform on $[0, 1]$, attains all upper endpoints at once. Hence $\text{cav } f = \sum_e a_e \text{cav } m_e$. For any other law P with these marginals, define its deficiency by

$$D_P(e) = \min_{i \in e} x_i - \mathbb{E}_P m_e(X) \geq 0.$$

If one law has $D_P(e) \geq \alpha H(m_e; x)$ for every nonempty support, then $H(f; x) \geq \alpha T(f; x)$. The law must be shared by all terms.

3 The universal upper bound

Define three global laws with the prescribed marginals:

- O : draw a common uniform U and, independently conditional on U , sample each coordinate with success probability

$$p_O(x_i, U) = \begin{cases} \frac{1}{2} \mathbf{1}_{\{U \leq 2x_i\}}, & x_i \leq 1/2, \\ 1 - \frac{1}{2} \mathbf{1}_{\{U \leq 2(1-x_i)\}}, & x_i > 1/2. \end{cases}$$

This is the folded orientation law used in the formal construction.

- I : draw independent Bernoulli coordinates of means x_i .
- B : call i low when $x_i \leq 1/2$. A low coordinate is $X_i = \mathbf{1}_{\{U \leq x_i\}}$. For each high coordinate, independently conditional on U , set $X_i = 0$ with probability $1/2$ when $U \leq 2(1 - x_i)$, and with probability zero otherwise.

For B , each high coordinate has failure probability $\frac{1}{2} 2(1 - x_i) = 1 - x_i$; all three laws preserve all means. Use the mixture

$$P = \frac{18}{31}O + \frac{6}{31}I + \frac{7}{31}B. \quad (1)$$

3.1 Cubic terms

Order the three means as $u \leq v \leq w$ and put $T = \min(u, 2 - v - w)$. Write D_O, D_I, D_B for the deficiencies. If $v \leq 1/2$, integrating the conditional product for the first two coordinates in O gives $\mathbb{E}_O[X_1 X_2] = u/2$. Thus $D_O \geq u/2$. Also $D_I = u(1 - vw) \geq u/2$. Here $T = u$, so $18D_O + 6D_I + 7D_B \geq 12T$.

If $u \leq 1/2 < v$, put $a = 1 - v$ and $b = 1 - w$, so $a \geq b \geq 0$. Integration of the conditional success probabilities gives

$$D_O = \frac{1}{2} \min(u, a) + \frac{1}{4} \min(u, b), \quad D_B = \frac{1}{2} \min(u, 2a) + \frac{1}{4} \min(u, 2b). \quad (2)$$

We claim $18D_O + 7D_B \geq 12 \min(u, a + b)$. If $a \leq u/2$, the left side is $16a + 8b \geq 12(a + b)$. If $b \geq u/2$, it is at least $18(3u/8) + 7(3u/4) = 12u$. In the remaining cases $b \leq u/2 \leq a$. For $a \leq u$ the left side is $9a + 8b + 7u/2$. When $a + b \leq u$, $3a + 4b \leq 7u/2$; when $a + b \geq u$, $9a + 8b \geq 8u + a \geq 17u/2$. Both give the claim. For $a \geq u$ the left side is $25u/2 + 8b \geq 12u$. The boundaries are included, and $D_I \geq 0$ completes the mixture estimate.

If $u > 1/2$, put $c = 1 - u$, $a = 1 - v$, $b = 1 - w$. Then $1/2 > c \geq a \geq b \geq 0$. Under B , the failure-union probability is $c + a/2 + b/4$, by integration over the three nested failure intervals. Direct integration for O gives the same deficiency:

$$D_O = D_B = a/2 + b/4 \geq 3(a + b)/8 \geq 3T/8.$$

Independence gives the stronger estimate $D_I \geq 7T/16$. Indeed, for $s = a + b \leq 1$ we have $ab \leq s^2/4$ and $D_I \geq u(s - s^2/4)$. If $s \leq u$, then $D_I/T \geq u(1 - u/4) \geq 7/16$. If $s \geq u$, then $D_I/T \geq s - s^2/4 \geq u - u^2/4 \geq 7/16$. These arguments are needed only for $T > 0$; at $T = 0$ nonnegativity suffices. Thus

$$18D_O + 6D_I + 7D_B \geq \left(25\frac{3}{8} + 6\frac{7}{16}\right)T = 12T.$$

3.2 Bilinear terms and boxes

For two means $u \leq v$, $T = \min(u, 1 - v)$ and $D_O = T/2$. If both are low, $D_I \geq T/2$. If just one is low, $D_B = \min(u, 2(1 - v))/2 \geq T/2$. If both are high, $D_B = T/2$ and $D_I = uT \geq T/2$. The mixture therefore captures at least $12T/31$ in every case. Constant and singleton terms have zero gap. Summing the common-law inequalities proves the cube bound.

For a nonnegative box, substitute $x_i = \ell_i + (u_i - \ell_i)y_i$. Each original positive monomial expands into nonnegative cube monomials of degree at most three. The sum of the expanded term widths is at least the original term width: taking a supremum of a sum is bounded by the sum of suprema, and taking an infimum has the reverse inequality. The full hull width is preserved by the affine change of coordinates. This remains true with zero-width coordinates. Apply the cube result to the expansion to prove the upper bound in Theorem 1.

4 Optimality of the fixed mixture

This section concerns uniform termwise guarantees for fixed mixtures of O, I, B , not the exact value of $R(3)$. Let their nonnegative weights be w, r, v with sum one. Suppose their mixture captures at least a fraction α of every bilinear and cubic monomial gap. Three tests give the following normalized deficiencies:

Means	Parameter	$(D_O, D_I, D_B)/T$ or its limit
$(1/2, 1/2)$	fixed	$(1/2, 1/2, 0)$
$(\varepsilon, 1 - \varepsilon^2, 1 - \varepsilon^2)$	$\varepsilon \downarrow 0$	$(3/8, 0, 3/4)$
$(u, 1 - u/2, 1 - u/2)$	$u \downarrow 1/2, u > 1/2$	$(3/8, 7/16, 3/8)$

In the second test take $0 < \varepsilon \leq 1/8$, so $T = 2\varepsilon^2$. The exact independent normalized deficiency is $\varepsilon - \varepsilon^3/2$. In the third test take $1/2 < u \leq 5/8$, which keeps all coordinates high and sorted; $T = u$, and the independent normalized deficiency is $u - u^2/4$. These are tests of the actual laws, with the threshold equality classified as low in the first test. It follows that

$$\alpha \leq \frac{1}{2} - \frac{v}{2}, \quad \alpha \leq \frac{3}{8} + \frac{3v}{8} - \frac{3r}{8}, \quad \alpha \leq \frac{3}{8} + \frac{r}{16}.$$

Average these inequalities with weights $3/31, 4/31, 24/31$. The coefficients of r and v cancel, giving $\alpha \leq 12/31$. Equation (1) attains that guarantee. This does not exclude a better law, coefficient-dependent weights, or an analysis that does not demand the same deficiency fraction separately for every term.

5 An analytic family for the lower bound

Let $m \geq 9$ be a multiple of nine. Use disjoint groups U, V, W of m variables each, with coordinate sums A, B, C . Write $E_k(G)$ for the elementary symmetric polynomial of degree k on a group G . Define

$$f_m = 2E_3(W) + 3BE_2(W) + 2mE_2(V) + \frac{5m}{3}AC + \frac{10m}{9}AB + mE_2(U). \quad (3)$$

Every included term has degree two or three and a positive integer coefficient. Set the group means to $1/4, 1/2, 3/4$, respectively, on the unit cube $[0, 1]^{3m}$.

5.1 A rational scalar certificate

Put

$$\begin{aligned} F(a, b, c) &= 6c^3 + 27bc^2 + 18b^2 + 30ac + 20ab + 9a^2, \\ L(a, b, c) &= 37a + \frac{79}{2}b + 38c - \frac{103}{3}, \quad \delta = \frac{901}{120000}. \end{aligned}$$

Lemma 2. For $(a, b, c) \in [0, 1]^3$, $F(a, b, c) \geq L(a, b, c) + \delta$.

Proof. Define

$$\begin{aligned} p(c) &= \frac{-972c^4 + 576c^3 + 1404c^2 - 768c + 101}{96}, \\ q(c) &= \frac{-78732c^4 + 212256c^3 - 203796c^2 + 82032c - 11275}{2976}. \end{aligned}$$

For $c \leq 3/10$, let $b_* = 13/24 - 3c^2/4$ and $g(c) = -49/6 + 30c - 15c^2 \leq 0$. The identity

$$F - L - p(c) = 9\left(a - 1 + \frac{10}{9}(b - b_*)\right)^2 + \frac{62}{9}(b - b_*)^2 + g(c)(a - 1)$$

proves $F - L \geq p(c)$ when $a \leq 1$. For every real a, b, c ,

$$\begin{aligned} F - L - q(c) &= 9\left(a + \frac{10}{9}b + \frac{30c-37}{18}\right)^2 \\ &\quad + \frac{62}{9}\left(b + \frac{9(27c^2 - 79/2) - 10(30c - 37)}{124}\right)^2. \end{aligned}$$

It remains to bound p on $[0, 3/10]$ and q on $[3/10, 1]$. For each row below, set $t = (c - \ell)/(u - \ell)$. The indicated polynomial equals $D^{-1} \sum_{i=0}^4 v_i \binom{4}{i} t^i (1 - t)^{4-i}$.

Polynomial	$[\ell, u]$	D	$(v_0, v_1, v_2, v_3, v_4)$
p	$[0, 1/5]$	60000	(63125, 39125, 20975, 9395, 4133)
p	$[1/5, 3/10]$	240000	(16532, 6008, 1802, 3788, 11597)
q	$[3/10, 1/2]$	7440000	(215357, 1285415, 1434125, 1250375, 1008125)
q	$[1/2, 3/4]$	190464	(25808, 18056, 7964, 9230, 15869)
q	$[3/4, 1]$	190464	(15869, 22508, 34520, 45920, 31040)

The Bernstein basis is nonnegative and sums to one. Every v_i/D is at least δ , proving the lemma by exact polynomial identities. $\square$

5.2 From counts to graph hulls

At a binary vertex, put $a = A/m$, $b = B/m$, $c = C/m$. Expansion gives

$$\frac{18}{m^3} f_m = F(a, b, c) - \frac{18c^2 + 27bc + 18b + 9a}{m} + \frac{12c}{m^2} \geq F(a, b, c) - \frac{72}{m}. \quad (4)$$

Every coupling of the prescribed individual marginals has expected normalized counts $(1/4, 1/2, 3/4)$. Taking expectations in Lemma 2 and (4) therefore yields

$$\frac{18}{m^3} \text{vex } f_m \geq \frac{139}{6} + \delta - \frac{72}{m}.$$

Counting the six support classes and using the individual endpoints gives

$$\begin{aligned} \frac{18}{m^3} \text{cav } f_m &= \frac{167}{4} - \frac{153}{4m} + \frac{9}{m^2}, \\ \frac{18}{m^3} T(f_m) &= \frac{161}{4} - \frac{135}{4m} + \frac{6}{m^2}. \end{aligned}$$

Only the $E_3(W)$ terms have nonzero individual lower endpoints. The actual hull gap is positive, and the preceding inequalities give

$$\frac{T(f_m)}{H(f_m)} \geq \frac{161/4 - 135/(4m) + 6/m^2}{223/12 - \delta + 135/(4m) + 9/m^2}. \quad (5)$$

Positivity of the hull gap follows, for example, by comparing the common upper law with independence: every nonlinear support here has means strictly between zero and one and a positive coefficient. As m tends to infinity through multiples of nine, the right side of (5) tends to

$$\frac{161/4}{223/12 - 901/120000} = \frac{1610000}{743033}.$$

Every member belongs to the class defining $R(3)$, so this proves the lower bound in Theorem 1. Dropping the slack gives the simpler bound $483/223$. Already the unrefined finite certificate at $m = 36$ exceeds two. No finite member is claimed to attain either limiting constant, and no limit of the actual ratios is asserted.

6 Exact finite examples

The formal project also verifies seven exact finite examples. For the two-group family, with both groups of size m , use

$$g_m = AE_2(W) + \frac{5m}{4} E_2(U), \quad x_U = 1/2, \quad x_W = 3/4.$$

For the three-group family, use group means $(1/4, 1/2, 3/4)$ and coefficients $\gamma = (\gamma_0, \dots, \gamma_5)$ on the six terms $(E_3(W), BE_2(W), E_2(V), AC, AB, E_2(U))$.

Family	Variables	Coefficients or parameter	Exact T/H
Two groups	8	$m = 4$	$27/16$
Two groups	16	$m = 8$	$21/11$
Two groups	24	$m = 12$	$99/50$
Two groups	32	$m = 16$	$135/67$
Three groups	18	$(2, 3, 9, 10, 7, 7)$	$20891/10411$
Three groups	24	$(2, 3, 13, 12, 8, 7)$	$6601/3225$
Three groups	192	$(2, 3, 120, 105, 70, 63)$	$7443345/3445256$

Each exact value is certified by an affine lower bound valid on every integer count vector, a probability law attaining that bound with all individual marginals preserved, and the common upper law. The last four ratios exceed two. These finite certificates are independent of the asymptotic argument above; their full data and kernel proofs are in the accompanying source distribution.

7 Formal verification and scope

The standalone project in `formal/` contains the dependency closure of the canonical proofs. Its `claim map` identifies declarations for the graph-hull interpretation, the three actual laws, scalar inequalities, all-box transfer, fixed-mixture obstruction, finite analytic family, limit and supremum, and finite witnesses. The `verification record` gives the pinned Lean and Mathlib versions, commands, module and declaration counts, axiom audit, kernel replay, and source fingerprints; source export identity and the paper build are checked separately. The standard axioms `propext`, `Classical.choice`, and `Quot.sound` are allowed; unfinished proofs and additional axioms are rejected.

General coefficient removal, homogeneous approximation of the supremum, equal-marginal classification, and the general two-group asymptotic family are separate results outside this paper; the finite two-group examples above do not depend on them.

References

- [1] James Luedtke, Mahdi Namazifar, and Jeff Lindereth. Some results on the strength of relaxations of multilinear functions. *Mathematical Programming*, 136(2):325–351, 2012. doi: 10.1007/s10107-012-0606-z. Author manuscript available at <https://jlindereth.github.io/papers/Luedtke-Namazifar-Lindereth-12-TR.pdf>.